INCH-POUND

AN929 Rev 14
w/AMENDMENT 1
26 September 2024
SUPERSEDING
AN929 Rev 14
29 August 2017

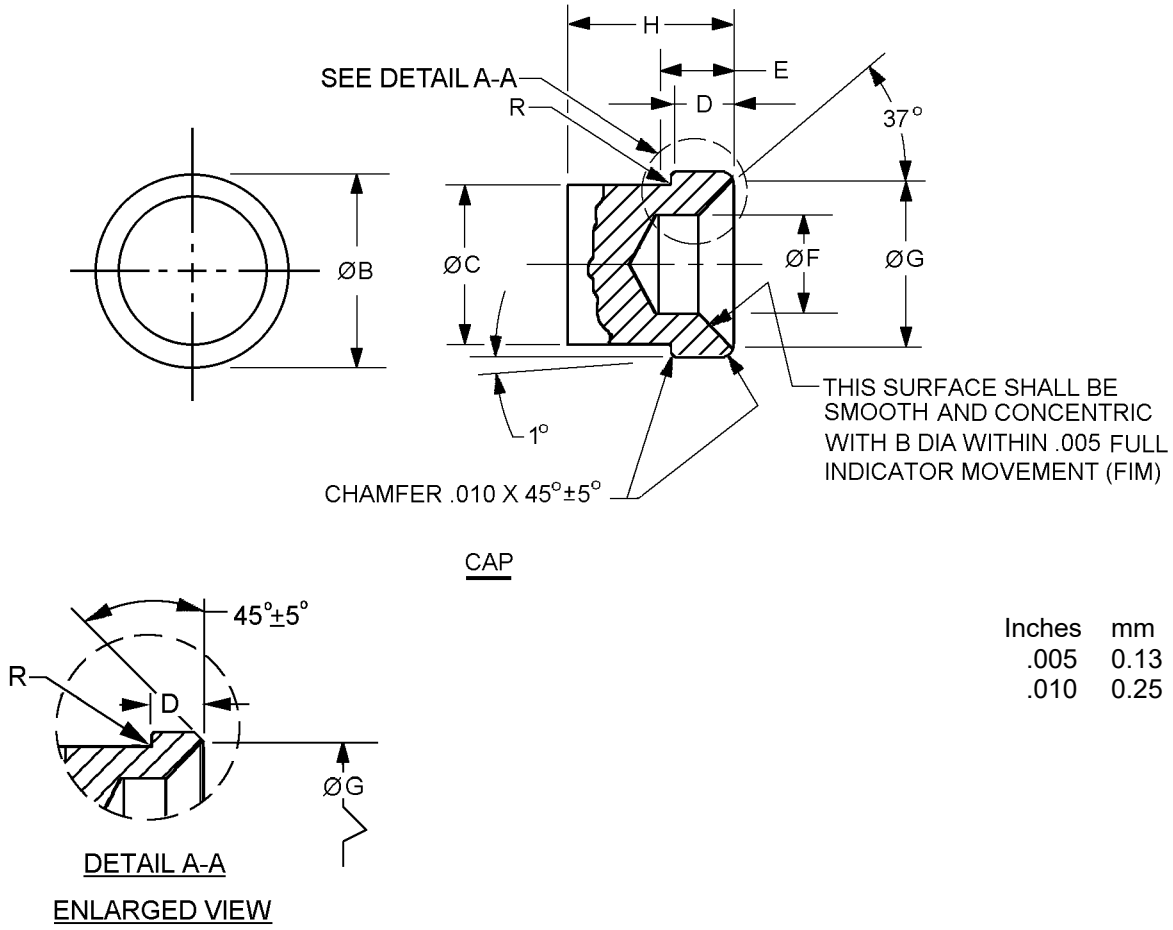
DETAIL SPECIFICATION SHEET

CAP INSERT, ASSEMBLY, PRESSURE SEAL, FLARED TUBE FITTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4841.

Cap dimensions and configuration, see figure 1. Cap and nut assembly, see figure 2.



Inches	mm
.005	0.13
.010	0.25

FIGURE 1. Cap dimensions and configuration.

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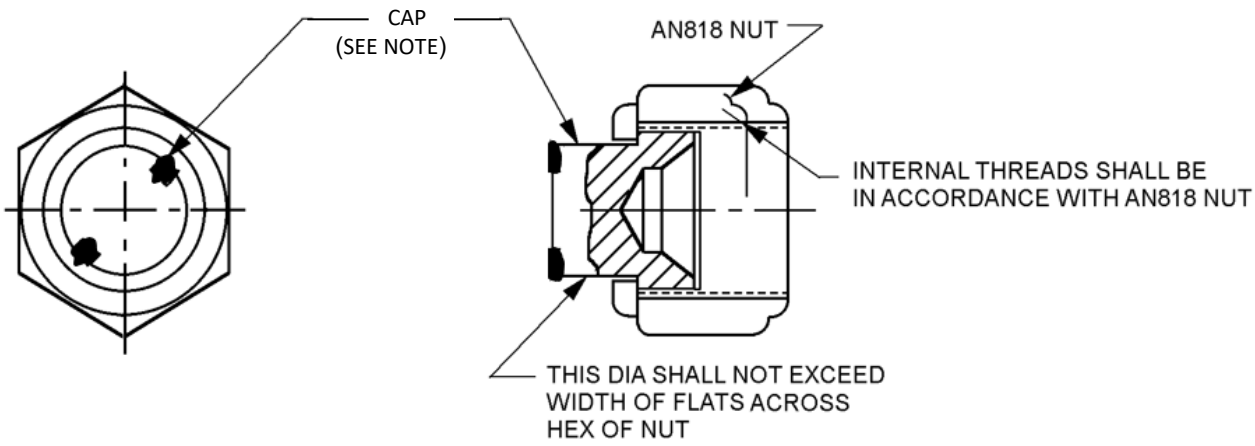
Dash number	Tube OD inches (mm)	B dia. inches (mm) +.000 -.003 (0.08)	C dia. inches (mm)	D inches (mm)	E inches (mm)
-2	.125 (3.18)	.263 (6.68)	.156 (3.96)	.156 (3.96)	.125 (3.18)
-3	.188 (4.78)	.325 (8.26)	.219 (5.56)	.219 (5.56)	
-4	.250 (6.35)	.378 (9.60)	.281 (7.14)	.203 (5.16)	
-5	.313 (7.95)	.441 (11.20)	.344 (8.74)	.219 (5.56)	.141 (3.58)
-6	.375 (9.53)	.503 (12.78)	.422 (10.72)	.266 (6.76)	.156 (3.96)
-8	.500 (12.70)	.683 (17.35)	.547 (13.89)	.313 (7.95)	.172 (4.37)
-10	.625 (15.88)	.799 (20.30)	.672 (17.07)	.281 (7.14)	.250 (6.35)
-12	.750 (19.05)	.974 (24.74)	.813 (20.65)		.313 (7.95)
-16	1.000 (25.40)	1.224 (31.09)	1.063 (27.00)	.297 (7.54)	
-20	1.250 (31.75)	1.536 (39.01)	1.328 (33.73)	.375 (9.53)	
-24	1.500 (38.10)	1.786 (45.36)	1.594 (40.49)	.406 (10.31)	.375 (9.53)
-28	1.750 (44.45)	2.161 (54.89)	1.875 (47.63)	.500 (12.70)	
-32	2.000 (50.80)	2.411 (61.24)	2.141 (54.38)	.578 (14.68)	.500 (12.70)
-40	2.500 (63.50)	2.911 (73.94)	2.641 (67.08)	.625 (15.88)	.688 (17.48)
-48	3.000 (76.20)	3.411 (86.64)	3.156 (80.16)	.688 (17.48)	

Dash number	F dia.	G dia. inches (mm) +.000 -.005 (0.13)	H inches (mm)	R radius max inches (mm)
-2	.094 (2.39)	.218 (5.54)	.531 (13.49)	.005 (0.13)
-3	.156 (3.96)	.250 (6.35)		
-4	.172 (4.37)	.300 (7.62)		
-5	.234 (5.94)	.360 (9.14)	.563 (14.30)	
-6	.297 (7.54)	.440 (11.18)	.625 (15.88)	.010 (0.25)
-8	.391 (9.93)	.575 (14.61)	.750 (19.05)	
-10	.438 (11.13)	.685 (17.40)		
-12	.563 (14.30)	.885 (22.48)	.875 (22.23)	
-16	.813 (20.65)	1.103 (28.02)		
-20	1.078 (27.38)	1.413 (35.89)	1.000 (25.40)	
-24	1.313 (33.35)	1.661 (42.19)	1.125 (28.58)	
-28	1.547 (39.29)	2.034 (51.66)	1.375 (34.93)	.015 (0.38)
-32	1.781 (45.24)	2.284 (58.01)	1.438 (36.53)	
-40	2.000 (50.80)	2.784 (70.71)	1.750 (44.45)	
-48	2.500 (63.50)	3.284 (83.41)	1.813 (46.05)	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Fittings shall be free of all burrs and slivers.
4. Unless otherwise specified, tolerances for three point decimals ±.010 inch (0.25 mm).
5. For design features purposes, this standard takes precedence over documents referenced herein.

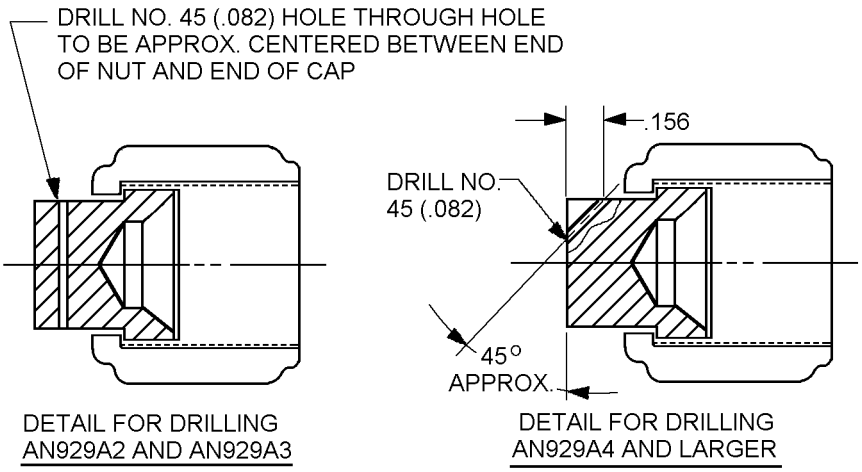
FIGURE 1. Cap dimensions and configuration - Continued.

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NOTE: Cap shall be movable and permanently secured to the nut to prevent assembly from coming apart, such as mechanical staking. The method used to prevent the movable cap from falling out of the nut is the manufacturer's option. Other methods may be used upon approval by the Preparing Activity. A physical and permanent retention method shall be present. See figure 4 for examples of approved methods.

FIGURE 2. Cap and nut assembly.



Inches	mm
.005	0.13
.082	2.08
.156	3.96

FIGURE 3. Cap assembly, safety chain configurations.

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REQUIREMENTS:

Dimensions and configurations: See figures 1, 2, and 3.

The Part or Identifying Number (PIN) represents a 2-piece assembly consisting of an AN818 nut and a permanently attached movable cap insert as specified herein and as shown on figure 2.

Material: See table I herein, procurement specification SAE-AS4841.

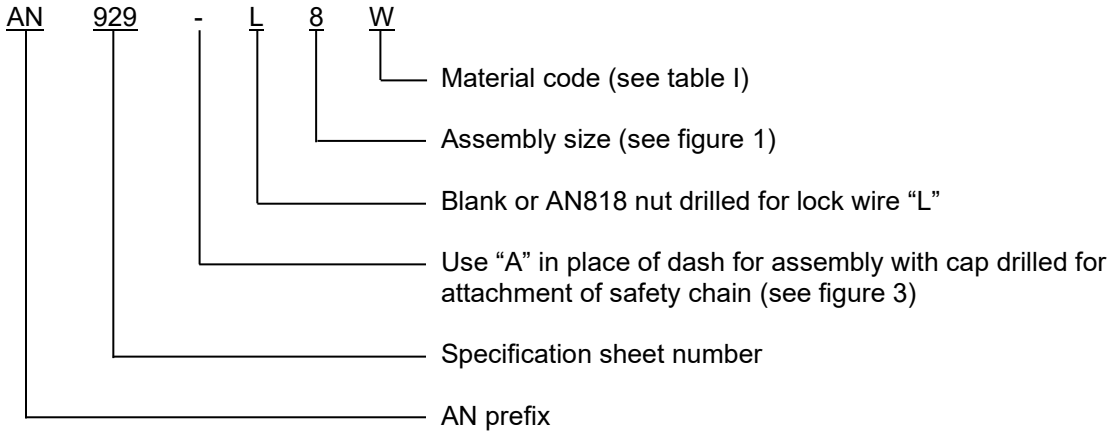
TABLE I. Material cap assembly.

Cap insert assembly AN929				Nut AN818 2/		
Material designator	Material	Alloy		Material designator	Material	Alloy
F	Steel	4130 or 4140		Blank	Steel	4130 or 4140
J	CRES	304		J	CRES	304
K	CRES	316		K	CRES	316
S	CRES	347		J or K	CRES	304 or 316
T 1/	Titanium	6Al-4V		T	Titanium	6Al-4V
W	Aluminum	7075-T73		W	Aluminum	7075-T73

1/ Not for use with oxygen systems.
2/ Material designator for manufacturers use only to determine proper AN818 nut to use for the cap insert assembly.

Finish: See SAE-AS4841.

PIN: The PIN consists of the AN prefix, specification sheet number, “A” in place of dash for assembly with cap drilled for attachment of safety chain, blank or “L” for nut drilled with lock wire, number for assembly size, and a letter for material designator.



PIN Examples:

AN929-8W identifies a cap assembly .500 inch (tube), 7075-T73 aluminum.
AN929-L8J identifies a cap assembly .500 inch (tube), AN818 nut drilled for lock wire, CRES 304.
AN929A8S identifies a cap assembly .500 inch (tube), cap drilled for attachment of safety chain, CRES 347.
AN929AL8J identifies a cap assembly .500 inch (tube), cap drilled for use with a safety chain, and AN818 nut drilled for lock wire, CRES 304.

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Supersession data: The blank material designator for aluminum alloys 2014 and 2024 in table II have been replaced by a “W” material designator for aluminum alloy 7075-T73 due to stress corrosion cracking aluminum alloys 2014 and 2024.

TABLE II. Supersession data.

Cancelled AN929 PIN		Replacement AN929 PIN
AN929-2		AN929-2W
AN929-L2		AN929-L2W
AN929A2		AN929A2W
AN929AL2		AN929AL2W
AN929-3		AN929-3W
AN929-L3		AN929-L3W
AN929A3		AN929A3W
AN929AL3		AN929AL3W
AN929-4		AN929-4W
AN929-L4		AN929-L4W
AN929A4		AN929A4W
AN929AL4		AN929AL4W
AN929-5		AN929-5W
AN929-L5		AN929-L5W
AN929A5		AN929A5W
AN929AL5		AN929AL5W
AN929-6		AN929-6W
AN929-L6		AN929-L6W
AN929A6		AN929A6W
AN929AL6		AN929AL6W
AN929-8		AN929-8W
AN929-L8		AN929-L8W
AN929A8		AN929A8W
N929AL8		AN929AL8W
AN929-10		AN929-10W
AN929-L10		AN929-L10W
AN929A10		AN929A10W
AN929AL10		AN929AL10W
AN929-12		AN929-12W
AN929-L12		AN929-L12W
AN929A12		AN929A12W
AN929AL12		AN929AL12W

Cancelled AN929 PIN		Replacement AN929 PIN
AN929-16		AN929-16W
AN929-L16		AN929-L16W
AN929A16		AN929A16W
AN929AL16		AN929AL16W
AN929-20		AN929-20W
AN929-L20		AN929-L20W
AN929A20		AN929A20W
AN929AL20		AN929AL20W
AN929-24		AN929-24W
AN929-L24		AN929-L24W
AN929A24		AN929A24W
AN929AL24		AN929AL24W
AN929-28		AN929-28W
AN929-L28		AN929-L28W
AN929A28		AN929A28W
AN929AL28		AN929AL28W
AN929-32		AN929-32W
AN929-L32		AN929-L32W
AN929A32		AN929A32W
AN929AL32		AN929AL32W
AN929-40		AN929-40W
AN929-L40		AN929-L40W
AN929A40		AN929A40W
AN929AL40		AN929AL40W
AN929-48		AN929-48W
AN929-L48		AN929-L48W
AN929A48		AN929A48W
AN929AL48		AN929AL48W

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FIGURE 4. Examples of approved methods of cap retention.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to SAE-AS4841, this document references AN818.

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force – 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2024-010)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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